

# WeatherNow – Weather API Project Report

## Real-Time Weather Data Visualization

### Project overview

WeatherNow is a weather-search web application that accepts a city name and displays current weather information such as temperature, feels-like temperature, humidity, wind speed, visibility, pressure, UV index, weather condition, and the latest update time.

**Important API identification:** The field names visible in the screenshots (for example *feelslike*, *humidity*, *wind\_kph*, *vis\_km*, *pressure\_mb*, *uv*, *last\_updated*, and *condition*) closely match the documented Current Weather response of WeatherAPI.com. WeatherAPI.com uses an API key and provides the current-weather endpoint */current.json*.

However, the screenshots alone cannot prove that the exact same API key from an earlier project was used. Also, the earlier project context available to me does not contain a stored WeatherNow API key or endpoint, so I would not claim the exact key is verified.

### Typical data flow

User enters city → WeatherNow sends a request to the weather API → API returns JSON → application extracts the required fields → UI displays the weather information.

API documentation reference: WeatherAPI.com

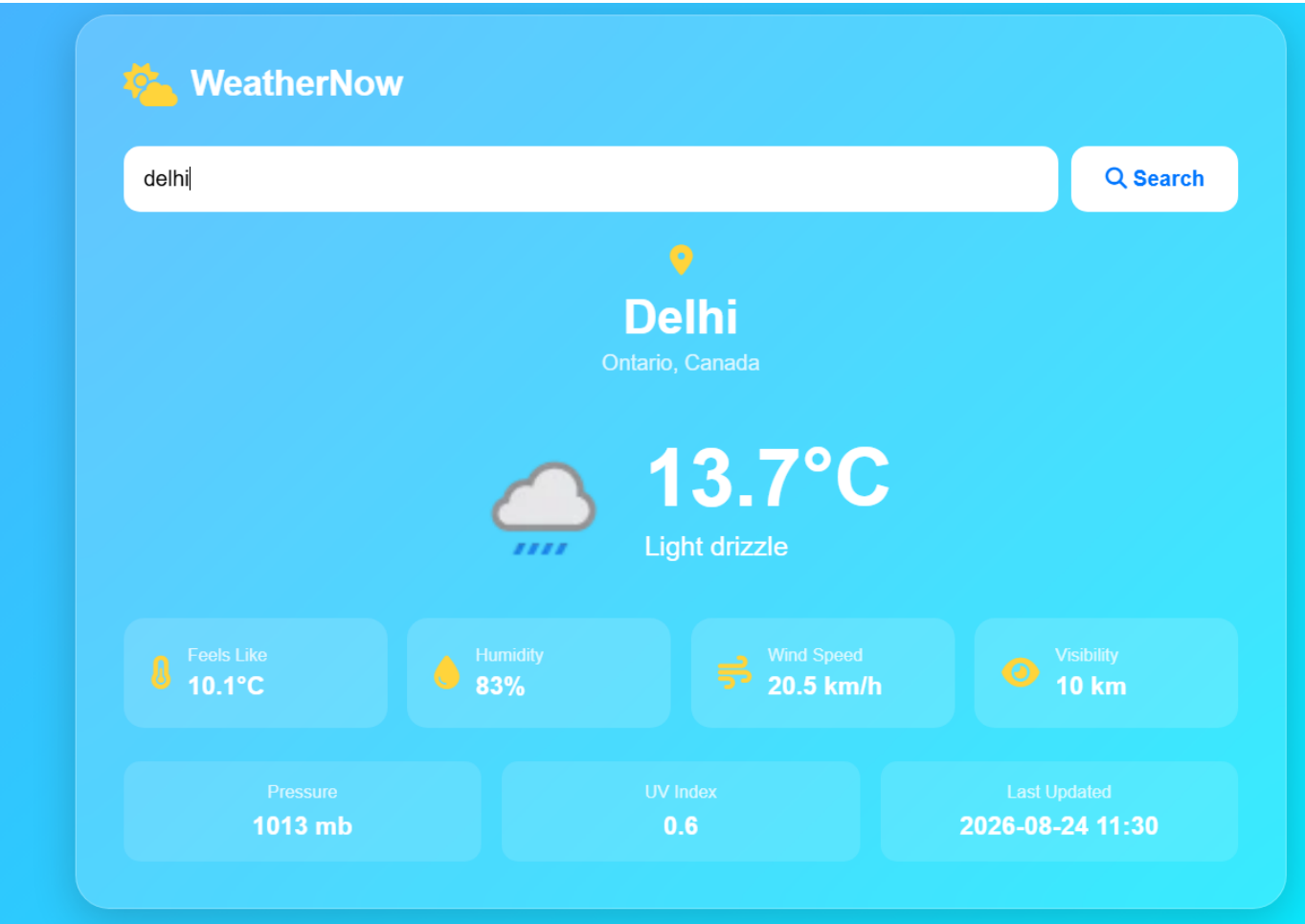
The official documentation describes API-key authentication and the Current Weather API fields used in the screenshots.

# Weather Data Summary

| City      | Country/Region       | Temperature | Condition          | Feels Like | Humidity | Wind      |
|-----------|----------------------|-------------|--------------------|------------|----------|-----------|
| Delhi     | Ontario, Canada      | 13.7 °C     | Light drizzle      | 10.1 °C    | 83%      | 20.5 km/h |
| Allahabad | Uttar Pradesh, India | 31.4 °C     | Overcast           | 38.7 °C    | 70%      | 5 km/h    |
| Mumbai    | Maharashtra, India   | 28.1 °C     | Patchy rain nearby | 32.2 °C    | 78%      | 24.8 km/h |

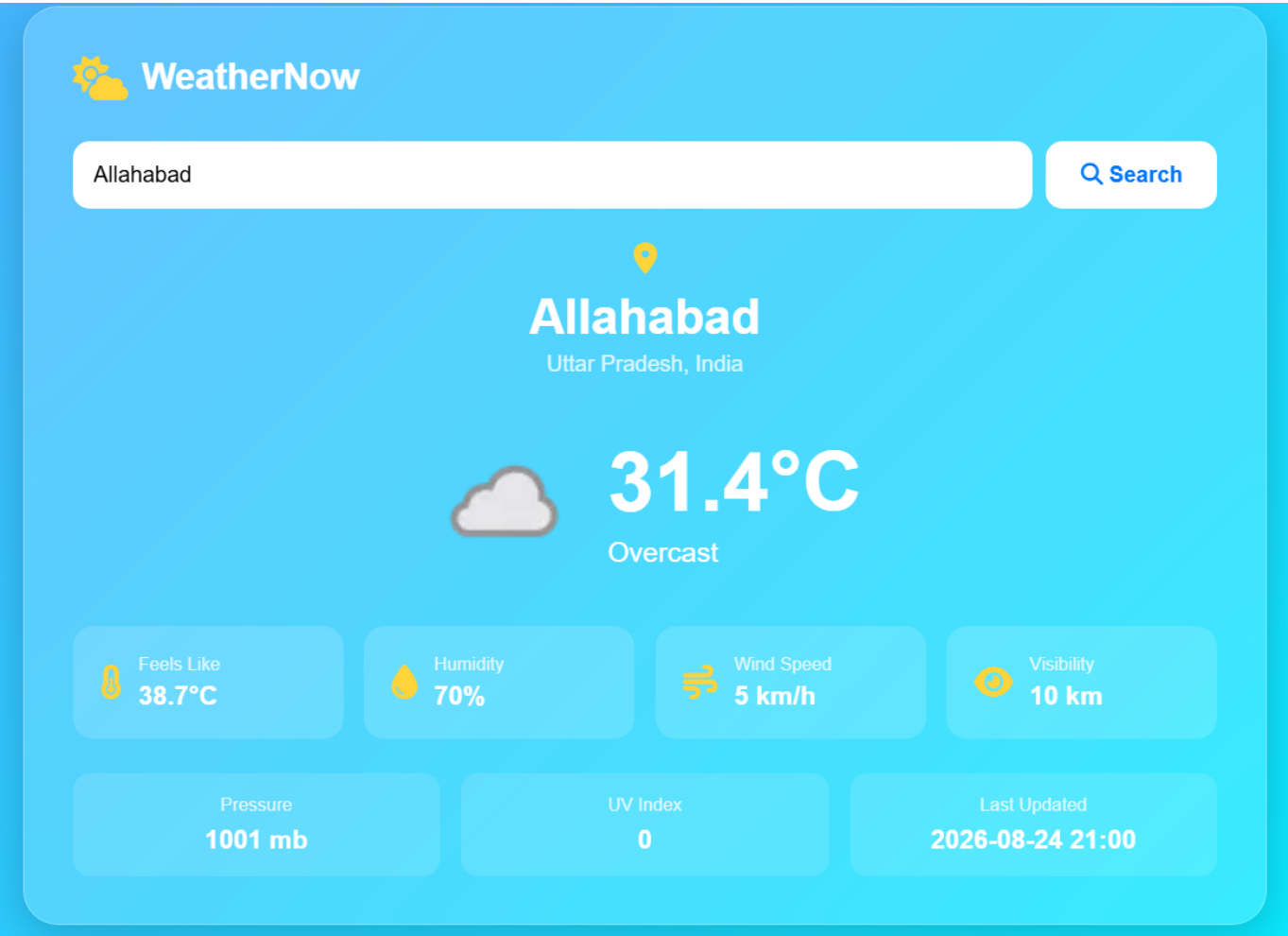
| City      | Visibility | Pressure | UV Index | Last Updated     |
|-----------|------------|----------|----------|------------------|
| Delhi     | 10 km      | 1013 mb  | 0.6      | 2026-08-24 11:30 |
| Allahabad | 10 km      | 1001 mb  | 0        | 2026-08-24 21:00 |
| Mumbai    | 10 km      | 1009 mb  | 0        | 2026-08-24 21:00 |

# Delhi – Ontario, Canada



Screenshot captured from the WeatherNow application.

# Allahabad – Uttar Pradesh, India



Screenshot captured from the WeatherNow application.

# Mumbai – Maharashtra, India



Screenshot captured from the WeatherNow application.

# How the Weather API data is used

- 1. Location search:** The user enters a city such as Mumbai, Allahabad, or Delhi in the search box.
- 2. API request:** The application sends the city and API key to the weather service. For WeatherAPI.com, the documented current-weather method is `/current.json` and authentication uses the `key` parameter.
- 3. JSON response:** The API returns structured location and current-weather objects. The current object contains fields such as temperature, condition, wind speed, pressure, humidity, feels-like temperature, visibility, UV index, and last-updated time.
- 4. Frontend display:** WeatherNow maps those JSON fields into the cards and values shown in the screenshots.
- 5. Dynamic search:** Searching another city causes the application to request new data, which is why the displayed location and weather values change.

## Key fields explained

**Temperature:** Current air temperature in Celsius.

**Feels Like:** The apparent temperature, which can differ from the actual air temperature.

**Humidity:** Relative humidity expressed as a percentage.

**Wind Speed:** Wind speed, displayed here in km/h.

**Visibility:** Approximate visibility distance in kilometres.

**Pressure:** Atmospheric pressure in millibars (mb).

**UV Index:** A measure of ultraviolet radiation intensity.

**Last Updated:** The local time associated with the latest weather observation returned by the API.

## Conclusion

The screenshots demonstrate a working real-time weather dashboard. Their data fields strongly correspond to WeatherAPI.com's documented current-weather response format. To verify the exact API key used by your original WeatherNow project, check the project's API request code or environment variable (for example, an `API_KEY` variable) rather than exposing the key itself in the report.